

# Janova

## Project Management SaaS

*From planning to progress, one place to make projects feel manageable.*

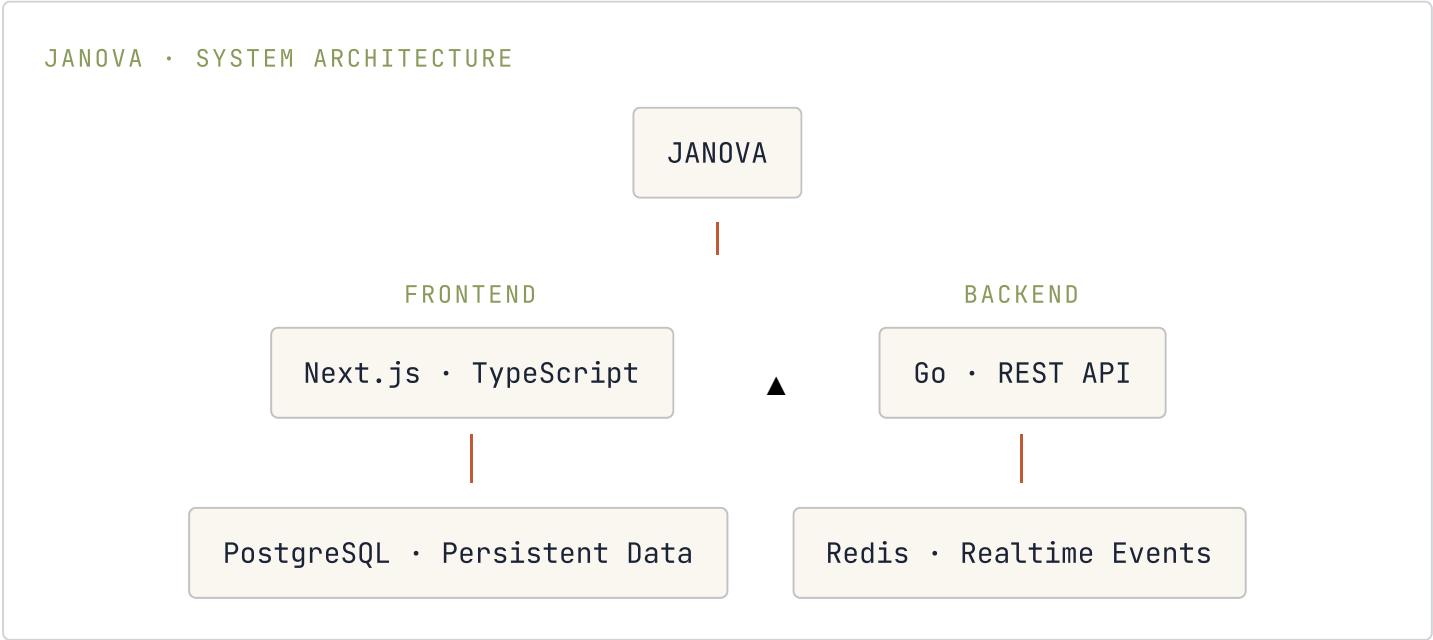
STORY

## The Story

Janova is a project management platform designed around the complete lifecycle of a project. Instead of treating project management as a collection of disconnected tools, Janova brings planning, budgeting, tasks, timelines, progress, collaboration, and files into one system.

SYSTEM

## Architecture



## STACK

# Tech Stack

## FRONTEND

Next.js

TypeScript

Interactive Kanban

Drag & drop

Realtime UI updates

## BACKEND

Go

REST API

Authentication

Authorization

Multi-tenant architecture

Project and company scoping

## DATA

PostgreSQL

sqlc

pgx

Structured relational data

## INFRASTRUCTURE

Docker

Redis

WebSocket / realtime architecture

## CONTRIBUTION

# What I Did

- Shaped the product end to end: user problem → product design → application → API → database → realtime system → deployment.
- Designed multi-tenant organizational scoping so projects and companies stay isolated.
- Built the interactive Kanban with drag & drop and realtime UI updates.
- Architected the Go REST API with authentication and authorization.

## PORTFOLIO STORY

*How do you turn a messy project workflow into software that people can actually use?*

## DEEP DIVE

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# FaS-Track Subsoil

Realtime Operational Software

*Turning field activity into software people can understand and act on.*

STORY

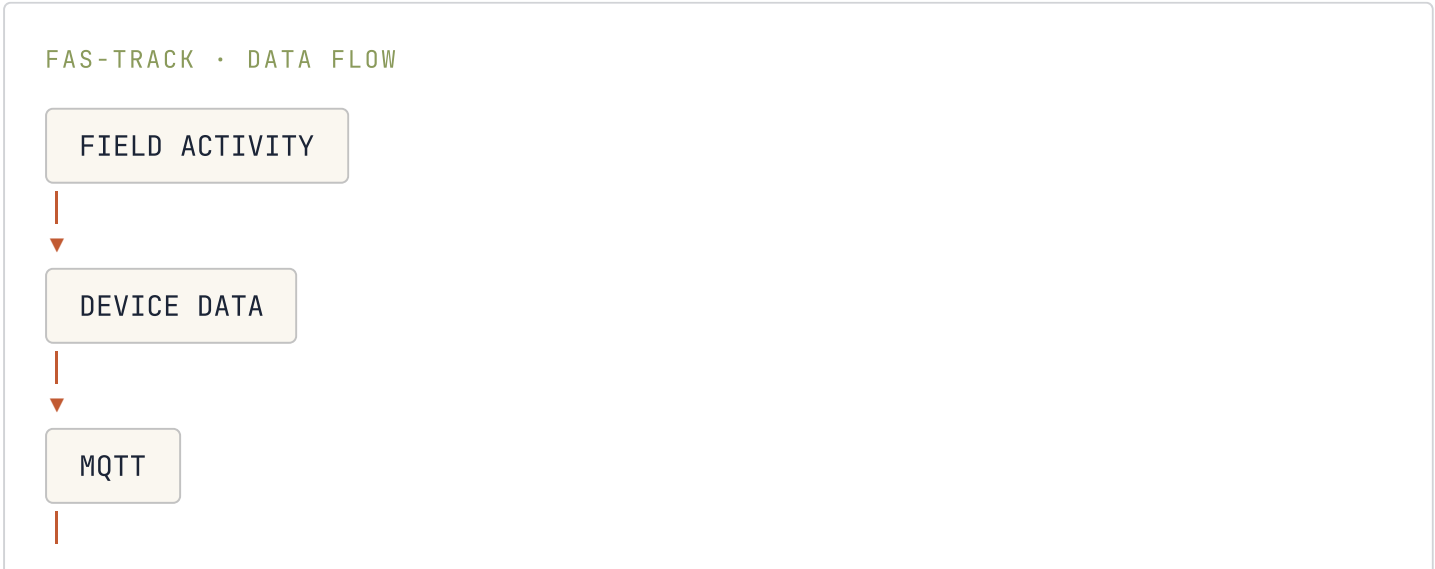
## The Story

FaS-Track Subsoil is an operational monitoring system developed for field machinery and subsoil operations.

Although the system involves connected devices, the focus here is a software and realtime data problem: machine activity and geospatial data flowing from the field into dashboards people act on.

SYSTEM

## Architecture





STACK

# Tech Stack

BACKEND

- NestJS
- TypeORM
- REST API
- MQTT
- WebSocket
- Redis

DATA

- PostgreSQL
- PostGIS
- Geospatial queries
- Telemetry data

FRONTEND

- React
- Vite
- Operational dashboards

INFRASTRUCTURE

- Docker
- Nginx
- PM2
- Linux

CONTRIBUTION

# What I Did

- Developed backend functionality for telemetry data.

- Worked with MQTT-based device communication.
- Designed and implemented data processing flows.
- Worked with PostgreSQL and PostGIS.
- Developed realtime application functionality.
- Built and improved operational dashboards.
- Worked with machine activity and geospatial data.
- Helped digitize operational processes.
- Worked across the system from incoming data to user-facing application.

#### KEY RESULT • GPS ACCURACY

**> 15 m error → < 5 m error**

#### PORTFOLIO STORY

*How do you turn continuously changing real-world data into software that people can actually understand?*

#### DEEP DIVE

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# BLE IoT Configuration App

Mobile Application × Connected Devices

*Making device configuration feel like using an app instead of debugging hardware.*

## STORY

# The Story

A Flutter mobile application for configuring IoT devices through Bluetooth Low Energy.

The important story is not the ESP32 itself — it is how a complicated device configuration process can become a simple mobile experience.

## SYSTEM

# Architecture

### BLE • USER FLOW

OPEN APP

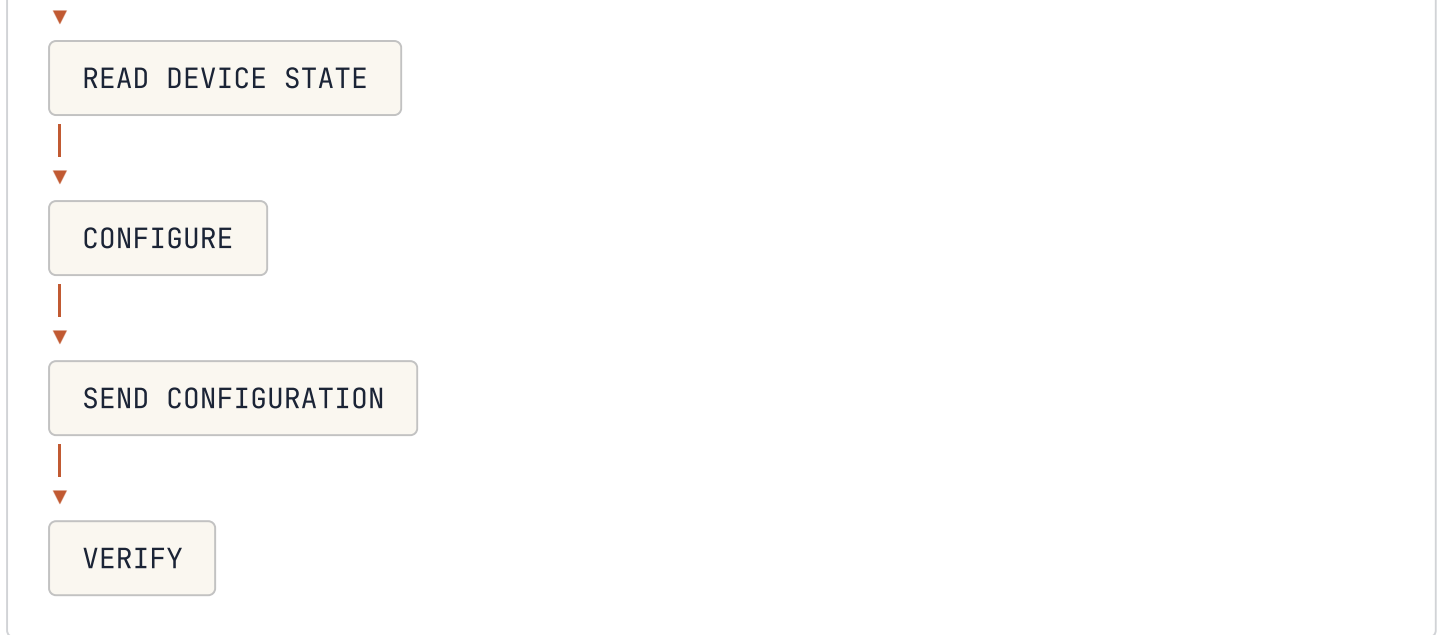


DISCOVER DEVICE



CONNECT VIA BLE





STACK

# Tech Stack

MOBILE

Flutter   FlutterBlue   BLE   UUID-based communication

DEVICE

ESP32 BLE Server   BLE Services   BLE Characteristics

CONFIGURATION

Wi-Fi SSID   Wi-Fi password   Endpoint URL   Device state   BLE serial monitoring

CONTRIBUTION

# What I Did

- Handled asynchronous BLE communication and device state gracefully: disconnected → scanning → connecting → connected → discovering services → ready → writing → verifying.
- Discovered services and characteristics via UUID-based protocols.
- Read device state and sent configuration reliably, with verification.



*How can a complicated device configuration process become a simple mobile experience?*

DEEP DIVE

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# Smart Halter

Turning Sensor Data Into Useful Information

*Raw data is not the product. Useful information is.*

STORY

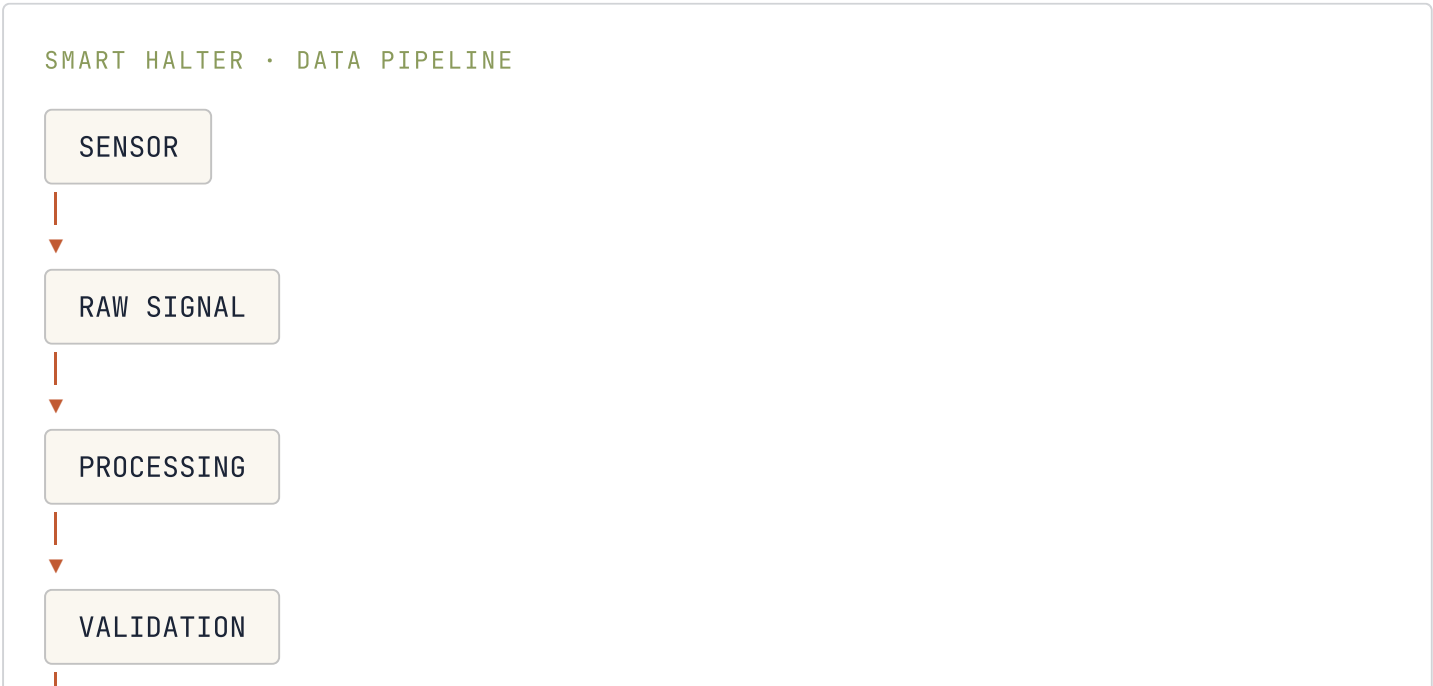
## The Story

Smart Halter improves the accuracy and usability of telemetry collected from a horse health monitoring system.

The system deals with measurements including heart rate, respiration, temperature, and SpO<sub>2</sub> — then turns those raw signals into information an application can act on.

SYSTEM

## Architecture





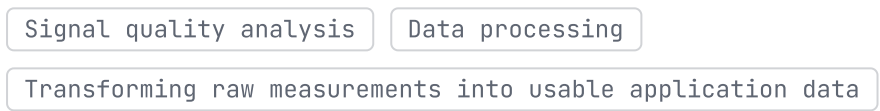
#### STACK

# Tech Stack

#### PIPELINE



#### SOFTWARE



#### CONTRIBUTION

# What I Did

- Improved sensor accuracy and usability of telemetry.
- Designed signal validation so bad readings are rejected before they reach the application.
- Worked across data processing, signal quality, and sensor integration.

KEY RESULT • SENSOR ACCURACY

**raw sensor output → ~60% improvement in sensor accuracy**

PORTFOLIO STORY

*How do you make imperfect real-world data useful?*

## DEEP DIVE

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